Lightweight Multi-Agent Fraud Detection Framework for Digital Payments in Nepal

[Aaroshree Gautam]
[Bachelor of Data Science Student, Kathmandu University]
aaroshiegautam@gmail.com

Abstract

The rapid expansion of digital payment platforms in Nepal—driven by services such as eSewa and IME Pay—has significantly increased exposure to financial fraud, yet fraud detection systems have not evolved at a comparable pace. According to Nepal Rastra Bank, mobile banking transactions grew by over 40% year-on-year in 2024, accelerating adoption even in semi-urban and rural areas. Existing approaches remain largely reactive, relying on static rule-based systems or isolated machine learning models that fail to capture the varying risk profiles associated with remittance, POS, and QR-based transactions.

This paper proposes a lightweight multi-agent fraud detection framework tailored to the constraints of Nepal's digital payment ecosystem. The system consists of five cooperative agents—Velocity, Geo, Behavior (Isolation Forest), Graph-inspired Risk, and Synthesis—each responsible for extracting complementary fraud signals from transaction streams. The central contribution is a context-aware synthesis mechanism that dynamically adjusts agent weight vectors and decision thresholds based on transaction type, enabling adaptive and proportionate responses in real time.

A multi-tier decision structure governs outcomes: transaction approval, single OTP verification, dual OTP authentication, or transaction blocking. The framework incorporates key robustness mechanisms such as cold-start user handling, SIM-swap attack mitigation through adaptive authentication escalation, and explicit false-positive reduction strategies to minimize legitimate transaction disruption.

Evaluated on a domain-informed synthetic dataset modeled on Nepal's digital payment behavior, the proposed system demonstrates interpretable, low-latency decision-making suitable for integration into existing banking infrastructure. By prioritizing transparency, contextual awareness, and deployability over black-box optimization, the framework offers a practical fraud detection approach aligned with the operational realities of emerging economies.

1. Introduction

The digital payments landscape in Nepal has undergone a rapid transformation over the past decade. Mobile wallet platforms such as eSewa and IME Pay now serve millions of active users, while the

widespread adoption of QR-based payments—accelerated by Nepal Rastra Bank (NRB) interoperability initiatives—has extended digital transactions into everyday commerce across urban and rural regions. According to Nepal Rastra Bank, mobile banking transactions grew by over 40% year-on-year in 2024, reflecting a structural shift away from cash-based systems toward inclusive digital finance.

However, this rapid expansion has not been matched by equivalent advancements in fraud detection infrastructure. The accessibility of digital payments—enabled by instant onboarding, low-friction transfers, and QR-based simplicity—has simultaneously expanded the attack surface for financial fraud. Common threats in Nepal's ecosystem include velocity-based attacks (rapid sequential transactions), location anomalies (geographically inconsistent activity), SIM-swap fraud (OTP interception via compromised mobile numbers), and exploitation of low-trust or newly created merchant accounts. These risks are further amplified by the heterogeneous nature of transactions, where remittance transfers, POS payments, and QR transactions exhibit fundamentally different behavioral and risk characteristics.

Existing fraud detection systems deployed within Nepali financial institutions typically fall into two categories. The first consists of static rule-based engines that apply fixed thresholds for anomaly detection. While interpretable and computationally efficient, these systems are rigid and prone to high false-positive rates when applied uniformly across diverse transaction contexts. The second category consists of isolated machine learning models trained on historical data. Although more adaptive, these approaches often lack contextual awareness, struggle with cold-start users who have no behavioral history, and operate as opaque black-box systems that are difficult to audit in regulated financial environments.

Neither approach adequately captures the layered and context-dependent nature of fraud in heterogeneous digital payment ecosystems.

To address this gap, this paper proposes a lightweight multi-agent fraud detection framework tailored to the constraints of Nepal's digital payment infrastructure. The system comprises five cooperative agents—Velocity, Geo, Behavior, Graph-inspired Risk, and Synthesis—each responsible for extracting a distinct category of fraud signal from transaction streams. A context-aware synthesis mechanism dynamically adjusts agent weight vectors based on transaction type (remittance, POS, or QR), enabling adaptive risk evaluation aligned with the underlying risk profile of each context. Final decisions are executed through a multi-tier OTP interlock system, which provides graduated responses ranging from transaction approval to single or dual OTP verification, and ultimately transaction blocking.

The framework is designed around three practical constraints often underemphasized in academic fraud detection systems:

- Deployability: low computational overhead, sub-second decision latency, and compatibility with existing banking infrastructure.

- Interpretability: transparent, rule-guided synthesis logic that can be audited and tuned without retraining complex models.
- Robustness: explicit mechanisms for handling cold-start users, mitigating SIM-swap attacks, and minimizing false positives to preserve customer experience.

The remainder of this paper is organized as follows. Section 2 reviews related work in fraud detection and multi-agent systems. Section 3 presents the system architecture and interaction model. Section 4 details the methodology of each agent and the synthesis mechanism. Section 5 discusses robustness strategies. Section 6 outlines the experimental setup and synthetic dataset design. Section 7 presents evaluation results and illustrative scenarios. Section 8 discusses deployment considerations, and Section 9 concludes with future directions.

2. Related Work

2.1 Rule-Based Fraud Detection Systems

Early fraud detection systems in banking relied predominantly on static rule engines—hand-crafted threshold conditions applied to transaction attributes such as amount, frequency, and geography. While interpretable and computationally inexpensive, these systems suffer from well-documented limitations: they require continuous manual maintenance, fail to generalize across evolving fraud patterns, and produce high false positive rates when applied uniformly across heterogeneous transaction types [1]. In emerging markets, where transaction behavior is highly variable and ground-truth fraud labels are scarce, rule-based systems remain common due to their low infrastructure requirements—but their rigidity makes them increasingly inadequate as digital payment volumes scale.

2.2 Machine Learning Approaches to Fraud Detection

The application of machine learning to fraud detection has been extensively studied. Supervised approaches—including logistic regression, random forests, and gradient boosting—have demonstrated strong performance on labeled datasets such as the IEEE-CIS Fraud Detection benchmark and the PaySim simulation framework [2, 3]. However, supervised methods are fundamentally constrained by the availability of labeled fraud examples, which are rare, expensive to obtain, and often unavailable in nascent digital payment ecosystems.

Unsupervised anomaly detection methods offer a practical alternative. Isolation Forest, introduced by Liu et al. [4], detects anomalies by isolating observations through random partitioning—anomalous transactions, being statistically rare and distinct, require fewer partitions to isolate. Its computational efficiency, resistance to the curse of dimensionality, and suitability for unlabeled data make it particularly well-suited for deployment in low-resource environments. Autoencoders and Local Outlier Factor (LOF) have also been explored for transaction anomaly detection [5], though their inference latency and interpretability trade-offs make them less appropriate for real-time banking applications.

A persistent limitation across these approaches is their treatment of all transactions as contextually equivalent—a remittance transfer and a QR micro-payment are evaluated through the same model with the same decision boundary, despite carrying fundamentally different risk semantics.

2.3 Multi-Agent Systems in Financial Risk

Multi-agent architectures have been applied in financial systems for portfolio management, market simulation, and risk assessment [6]. The core advantage of agent decomposition is separation of concerns: each agent specializes in a narrow signal domain, making the overall system more modular, interpretable, and maintainable than a monolithic model. In fraud detection specifically, agent-based approaches have been proposed for credit card fraud [7] and insurance claim analysis, though their adoption in real-time payment systems—particularly in emerging markets—remains limited.

Prior multi-agent fraud detection frameworks have largely focused on agent communication and consensus rather than context-sensitive signal weighting. Our framework differs fundamentally: the Synthesis Agent does not simply aggregate outputs—it applies transaction-type-specific weight vectors, treating context as a first-class decision variable.

2.4 Fraud Detection in Emerging Digital Payment Ecosystems

Research specifically addressing fraud detection in South Asian or similarly structured digital payment ecosystems remains sparse. Most published work assumes data-rich environments with large labeled datasets, mature regulatory frameworks, and reliable device fingerprinting infrastructure—conditions that do not reflect Nepal's current landscape. Closest in spirit are studies on mobile money fraud in Sub-Saharan Africa [8], which similarly highlight the tension between security rigor and user experience in low-infrastructure contexts, and the importance of false positive minimization in markets where customer trust is fragile and churn is high.

Our work contributes to this underexplored space by proposing a framework explicitly designed around the constraints of emerging economies: synthetic data compatibility, rule-based interpretability, cold-start resilience, and sub-second latency without GPU dependency.

2.5 Positioning

Table 1: Comparison of fraud detection approaches across key deployment criteria.

Approach	Contextual Awareness	Interpretable	Cold-Start Handling	Low Latency
Rule-based engines	No	Yes	No	Yes
Isolated ML models	No	No	No	Yes
Prior multi-agent systems	Partial	Partial	No	Partial
This work	Yes	Yes	Yes	Yes

3. System Architecture

3.1 Agent Descriptions

Velocity Agent

Monitors transaction frequency over rolling time windows to detect abnormal spikes indicative of account takeover or automated fraud patterns. Output: velocity risk score $S_{velocity} \in [0, 1]$.

Geo Agent

Evaluates geographic consistency using historical user location patterns and device metadata. Flags include sudden location shifts and device-location mismatches. Output: geo risk score $S_{geo} \in [0, 1]$.

Behavior Agent

Applies an Isolation Forest model on behavioral features such as transaction amount, time of day, transaction type, merchant category, and derived frequency metrics to detect deviations from normal user behavior. Output: anomaly score $S_{behavior} \in [0, 1]$.

Graph-inspired Risk Agent

Approximates merchant-level risk using three signals: (i) novelty of user–merchant interaction, (ii) merchant transaction frequency, and (iii) merchant connectivity based on unique user interactions. These three signals correspond to graph-theoretic concepts — interaction novelty encodes edge existence, merchant frequency encodes node degree, and unique user count encodes a first-order approximation of betweenness centrality — without requiring persistent graph storage or retraining cycles. Together, they provide a lightweight graph-inspired representation of merchant trustworthiness that preserves relational signal at a fraction of the computational cost of full graph neural networks. Output: merchant risk score $S_{graph} \in [0, 1]$.

Synthesis Agent

Receives the four agent scores and a transaction context label, and computes a composite risk score R as a context-aware convex combination. Outputs the final score and delegates the decision to the OTP Interlock.

3.2 OTP Interlock

The OTP Interlock converts the composite risk score into one of four graded responses:

Table 2: OTP Interlock decision tiers.

Risk Tier	Action	Description
Low	Approve	Transaction proceeds without friction
Medium	Single OTP	Light verification for mild anomalies
High	Dual OTP	Strong verification for elevated risk

Risk Tier	Action	Description
Very High	Block	Transaction is halted

Thresholds are context-dependent rather than globally fixed. High-value remittance transactions trigger earlier escalation, while low-value QR payments tolerate higher risk scores before intervention. This ensures a balance between security and user experience across heterogeneous transaction types.

3.3 System Flow Summary

Transaction processing follows a structured pipeline:

1. Transaction received.
2. Context classification (remittance / POS / QR).
3. Parallel execution of five agents.
4. Synthesis Agent applies context-aware weighting.
5. OTP Interlock determines final action.

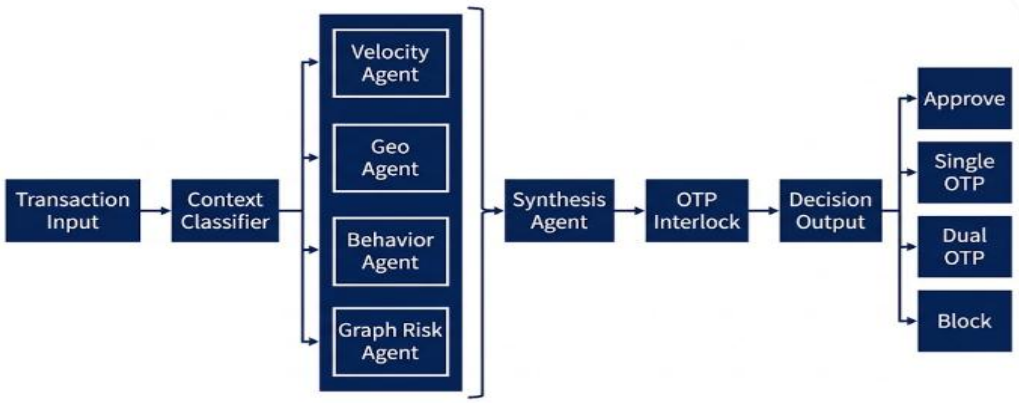


Figure 1: Multi-Agent Fraud Detection Pipeline

3.4 Design Rationale

Three key architectural decisions underpin the system design:

Full signal aggregation prior to synthesis ensures that no individual risk signal is prematurely discarded. Even weak individual anomalies can become significant when combined, improving detection robustness through emergent signal interactions.

Separation of context classification from synthesis ensures transparency and auditability. The applied weight vector can be explicitly traced for any transaction, enabling regulatory interpretability and post-hoc inspection of decision logic.

Graph-inspired approximation over full GNN models avoids the computational and infrastructural overhead of graph neural networks while preserving essential relational signals such as merchant rarity and interaction sparsity. This trade-off enables sub-second inference suitable for low-resource financial environments.

4. Methodology

This section details the internal logic of each agent and the synthesis mechanism that combines their outputs into a final risk decision.

4.1 Velocity Agent

The Velocity Agent monitors transaction frequency over three fixed rolling time windows, each targeting a distinct class of fraud behavior:

Table 3: Velocity Agent time windows.

Window	Duration	Purpose
Short	1 minute	Detects burst attacks and rapid-fire automated fraud
Medium	1 hour	Captures moderate velocity anomalies
Long	24 hours	Identifies daily volume abuse patterns

Each window maintains an independent transaction count for the user. When the count exceeds the predefined threshold for that window, a binary flag is raised. The velocity risk score is computed as a weighted sum of individual window violations, augmented by an interaction term that amplifies the score when multiple windows are violated simultaneously:

```
S_velocity = min(1,  v_s * f_s  +  v_m * f_m  +  v_l * f_l  +  λ *
                        f_s * f_m * f_l)
```

where f_s, f_m, f_l ∈ {0, 1} are binary violation flags for the short, medium, and long windows respectively; v_s, v_m, v_l are their associated severity weights; λ is an amplification constant active only when all three flags are simultaneously raised; and min(1, ·) caps the score within [0, 1].

The base computation is additive—each violated window contributes independently. The interaction term λ · f_s · f_m · f_l activates only under simultaneous multi-window violation, producing a score disproportionately higher than any single violation alone.

Intuition: A user triggering only the short window may represent a brief legitimate spike—for example, splitting a group bill. A user simultaneously violating all three windows is almost certainly engaged in systematic abuse. The interaction term captures this qualitative difference without discarding the independent signal value of each window.

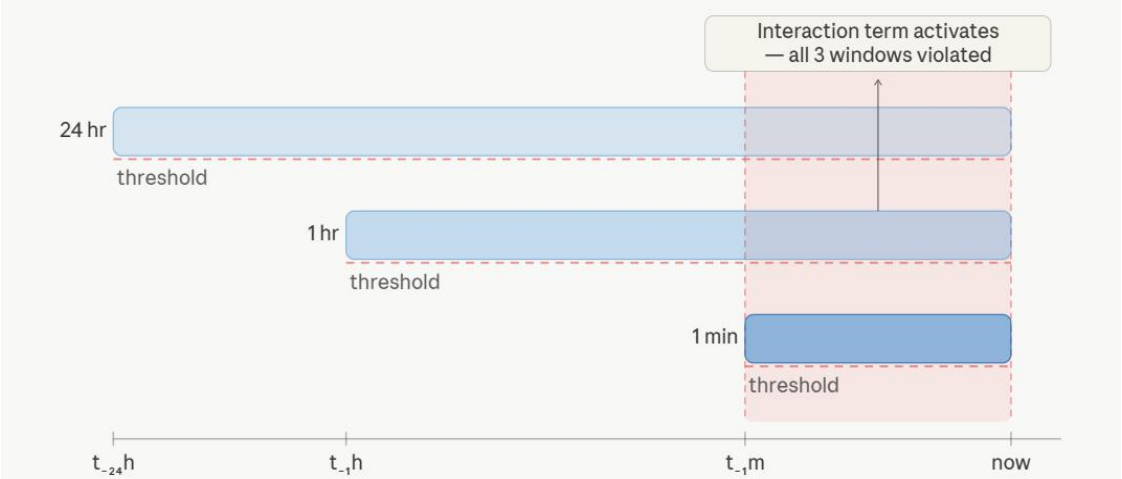


Figure 2: Velocity Agent — Rolling Window Violation Detection

4.2 Geo Agent

The Geo Agent evaluates the geographic plausibility of each transaction by comparing current transaction metadata against the user's established location history. Two primary signals are assessed:

Location Consistency Score: Measures the geographic distance between the current transaction origin and the user's most recent known locations. Large deviations—particularly those physically implausible given the time elapsed since the last transaction—receive high risk scores.

Device-Location Mismatch Flag: Detects inconsistencies between the device fingerprint associated with the transaction and the expected device for that location profile. A transaction from a known device in an unexpected location, or from an unknown device in a known location, each contribute independently to the geo risk score.

$$S_geo = \alpha \cdot \delta_location + (1 - \alpha) \cdot \delta_device$$

where $\delta_location \in [0, 1]$ is the normalized location deviation score, $\delta_device \in \{0, 1\}$ is the device mismatch binary flag, and α is a tunable parameter balancing the two signals (default $\alpha = 0.6$).

Intuition: In Nepal's context, a remittance transaction suddenly originating from a different district—or worse, a different country—without any prior travel history is a strong fraud indicator. The Geo Agent is specifically sensitive to this pattern.

4.3 Behavior Agent

The Behavior Agent applies an Isolation Forest model to detect statistical deviations from a user's established behavioral profile. Isolation Forest operates by randomly partitioning the feature space through a series of decision trees—anomalous observations, being rare and distinct, are isolated in fewer partitions and receive higher anomaly scores.

Table 4: Behavior Agent feature set.

Feature	Description

Feature	Description
Transaction amount	Absolute value of the transaction
Hour of day	Time-of-day encoding (0–23)
Transaction type	Categorical: remittance, POS, QR
Merchant category	Categorical: retail, utility, transfer, etc.
Frequency (short window)	Transaction count in last 1 minute
Frequency (medium window)	Transaction count in last 1 hour

The model is trained per-user on historical transaction records to establish a personalized behavioral baseline. At inference time, the anomaly score produced by Isolation Forest is normalized to [0, 1], where higher values indicate greater deviation from normal behavior.

Model configuration: Baseline Isolation Forest with default contamination parameter, intentionally kept simple to preserve low inference latency and avoid overfitting on limited synthetic data.

Intuition: If a user who consistently transacts small amounts at local merchants at midday suddenly initiates a large transfer to an unfamiliar merchant category at 3 AM, the Behavior Agent will flag this as anomalous—even if the velocity and geo signals appear normal.

4.4 Graph-inspired Risk Agent

The Graph-inspired Risk Agent estimates merchant-level trustworthiness using three lightweight signals that approximate the relational structure of a graph without requiring persistent graph storage or retraining cycles.

Let $S_{graph} \in [0, 1]$ denote the final merchant risk score, computed as the unweighted mean of three normalized component signals:

$$S_{graph} = (N + F + C) / 3$$

Signal 1 — Interaction Novelty (N)

Captures whether the current user has previously transacted with this merchant. Trust accumulates with repeated interactions via exponential decay:

$$N = 1.0 \quad \text{if } c = 0 \text{ (first-time interaction)}, \qquad e^{(-k \cdot c)} \quad \text{if } c > 0$$

where c is the count of prior user–merchant interactions and $k > 0$ is a decay constant controlling the rate of trust accumulation.

Signal 2 — Merchant Frequency (F)

Captures merchant rarity across the full transaction population. Rare merchants receive higher risk scores:

$$F = 1 - (t_m / t_{max})$$

where t_m is the total transaction count for merchant m and t_max is the maximum transaction count observed across all merchants.

Signal 3 — Merchant Connectivity (C)

Captures how many unique users have transacted with this merchant. Low-connectivity merchants are treated as less established:

$$C = 1 - (u_m / u_{max})$$

where u_m is the number of unique users who have transacted with merchant m and u_max is the maximum observed across all merchants.

Intuition: A merchant appearing for the first time (N = 1.0), with very few system-wide transactions (F ≈ 1.0), and only one known customer (C ≈ 1.0) receives S_graph ≈ 1.0—maximum risk. This captures new merchant exploitation fraud without requiring a full graph neural network.

4.5 Synthesis Agent

The Synthesis Agent receives four agent risk scores and the transaction context label, and computes a final composite risk score R as a convex combination of agent outputs:

$$R = w_1 \cdot S_{velocity} + w_2 \cdot S_{geo} + w_3 \cdot S_{behavior} + w_4 \cdot S_{graph}$$

where $w_i \geq 0$ for all i and $\Sigma w_i = 1.0$. Because each agent score $S_i \in [0, 1]$ and the weights form a convex combination, the composite score is guaranteed $R \in [0, 1]$ by construction—no post-hoc clipping or normalization is required.

Table 5: Context-specific weight vectors for the Synthesis Agent.

Agent	Remittance	POS	QR
Velocity (w1)	0.15	0.40	0.30
Geo (w2)	0.40	0.25	0.25
Behavior (w3)	0.30	0.20	0.25
Graph Risk (w4)	0.15	0.15	0.20
Sum	1.00	1.00	1.00

Weight Rationale:

Remittance—typically low-frequency, high-value periodic transfers. Velocity anomalies are less diagnostic given the inherently low transaction frequency of this context. Geographic anomalies are highly informative—a sudden change in sending location without travel history is a strong fraud signal. Geo therefore receives the highest weight (0.40).

POS—frequent, location-bound, time-sensitive transactions. Velocity is the most discriminative signal; an unusual number of POS transactions in a short window is the dominant fraud pattern in this context.

QR payments—low-value, high-frequency, spanning diverse merchant types. No single signal dominates, justifying a balanced distribution. Graph Risk receives a marginally elevated weight (0.20) reflecting the prevalence of QR-based merchant fraud through newly created or obscure merchant accounts.

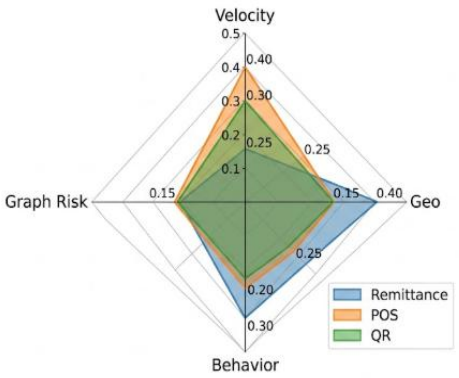


Figure 3: Context-Aware Weight Vectors by Transaction Type

4.6 OTP Interlock — Threshold Mapping

The composite score R is mapped to a decision action through context-specific threshold bands. Each transaction type maintains its own threshold table, reflecting its distinct risk tolerance.

Remittance (strictest thresholds — high value, high sensitivity)

R Range	Action
$R < 0.35$	Approve
$0.35 \leq R < 0.60$	Single OTP
$0.60 \leq R < 0.80$	Dual OTP
$R \geq 0.80$	Block

POS (moderate thresholds — frequent, lower individual risk)

R Range	Action
$R < 0.50$	Approve
$0.50 \leq R < 0.70$	Single OTP
$0.70 \leq R < 0.90$	Dual OTP
$R \geq 0.90$	Block

QR Payment (most tolerant thresholds — low value, high frequency)

R Range	Action

R Range	Action
$R < 0.55$	Approve
$0.55 \leq R < 0.75$	Single OTP
$0.75 \leq R < 0.90$	Dual OTP
$R \geq 0.90$	Block

All threshold values are heuristically defined based on domain risk sensitivity and user experience considerations. In a production environment, these can be calibrated empirically using historical fraud data and optimized against precision-recall trade-offs.

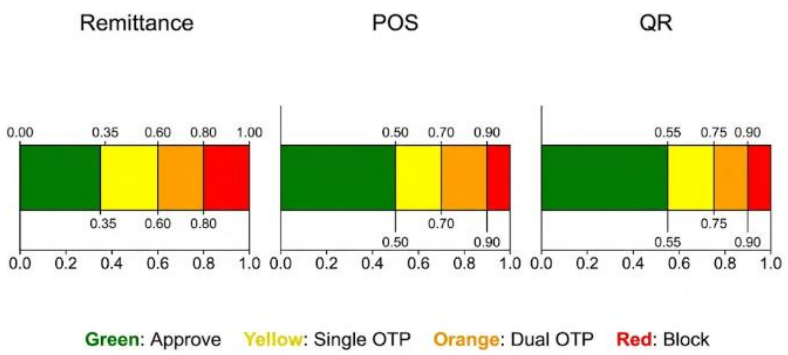


Figure 4: Context-Specific OTP Interlock Threshold Bands

5. Robustness Enhancements

A fraud detection framework is only as strong as its ability to handle edge cases that fall outside normal operating assumptions. This section addresses three robustness challenges explicitly designed into the framework: cold-start user handling, SIM-swap attack mitigation, and false positive reduction.

5.1 Cold-Start User Handling

The Problem: New users present a fundamental challenge for behavioral fraud detection—without transaction history, the Behavior Agent has no established baseline to detect deviations from, and the Graph-inspired Risk Agent has no prior user–merchant interaction records. A naive system would either flag all new user transactions as high risk (producing excessive false positives) or ignore the absence of history entirely (leaving a clear exploitation window).

The Approach: The framework handles cold-start users through a combination of conservative defaults and graceful degradation:

Default Risk Elevation: In the absence of a behavioral baseline, the Behavior Agent returns a moderate default anomaly score rather than zero. This reflects genuine uncertainty—the system does not know whether the user's behavior is normal or anomalous, and that uncertainty itself carries risk signal.

$$S_{behavior}(cold) = \beta \quad \text{where } \beta = 0.5$$

In our implementation, $\beta = 0.5$—the midpoint of the scoring range—representing maximum uncertainty about the user's behavioral profile. This ensures mild OTP friction is applied to new users without treating them as high-risk by default.

Graph Risk Elevation: First-time users interacting with any merchant receive $N = 1.0$ (maximum novelty score) by definition, naturally elevating S_{graph} for all cold-start transactions. No special handling is required—the novelty signal already encodes the absence of interaction history.

Velocity and Geo Bootstrapping: For new users, velocity thresholds are tightened by a cold-start multiplier $\gamma < 1$, reducing the number of transactions required to trigger a flag. The Geo Agent defaults to moderate risk for the first transaction since no location history exists to compare against.

Graduated Trust Accumulation: As a user completes verified transactions over time, their behavioral profile is incrementally updated and the cold-start defaults are phased out. After a configurable number of successful transactions—for example, five verified interactions—the system transitions the user to full profile-based evaluation.

Intuition: A new user is not assumed fraudulent, but is treated with appropriate caution. The system applies light friction (typically Single OTP) to early transactions and relaxes it as trust accumulates—mirroring the natural onboarding caution of any responsible financial institution.

5.2 SIM-Swap Attack Mitigation

The Problem: SIM-swap fraud occurs when an attacker convinces a mobile carrier to transfer a victim's phone number to a SIM card under the attacker's control. Once successful, the attacker can intercept OTP messages intended for the victim, bypassing standard single-factor OTP verification entirely. This attack is particularly relevant in Nepal's context, where OTP-based authentication is the primary—and often sole—second factor for digital payment authorization.

The Approach: The framework addresses SIM-swap risk through multi-signal detection and adaptive authentication escalation:

Geo-Device Divergence Detection: A SIM-swap attack frequently produces a characteristic signal pattern—the transaction originates from a familiar location but from an unrecognized device, or vice versa. The Geo Agent's device-location mismatch flag is specifically sensitive to this pattern. A high δ_{device} score combined with a moderate $\delta_{location}$ score produces an elevated S_{geo} that propagates into the composite risk score.

Behavioral Discontinuity: Following a SIM-swap, attackers typically initiate transactions that deviate significantly from the victim's established behavioral profile—different amounts, unusual

times, unfamiliar merchant categories. The Behavior Agent flags these deviations as anomalous, contributing an elevated S_behavior score independently of the geo signal.

Synthesis-Level Amplification: When both S_geo and S_behavior are simultaneously elevated—a pattern consistent with SIM-swap scenarios—the composite score R naturally reflects this through the weighted combination. For remittance transactions, where geo weight is highest (0.40), this amplification is most pronounced.

Adaptive Authentication Escalation: When the composite score places a transaction in the Dual OTP tier, the system requires verification through a channel independent of SMS. This directly neutralizes the SIM-swap attack vector, since the attacker controls the victim's SMS channel but not necessarily their registered email or authenticated app session. In Nepal's context, this escalation could be fulfilled through the bank's mobile banking app push notification or registered email.

Intuition: The framework does not require explicit SIM-swap detection as a separate module. Instead, SIM-swap attacks naturally produce a combination of geo anomaly and behavioral discontinuity signals that the existing agents detect independently. The Dual OTP response then escalates authentication to a channel the attacker cannot intercept.

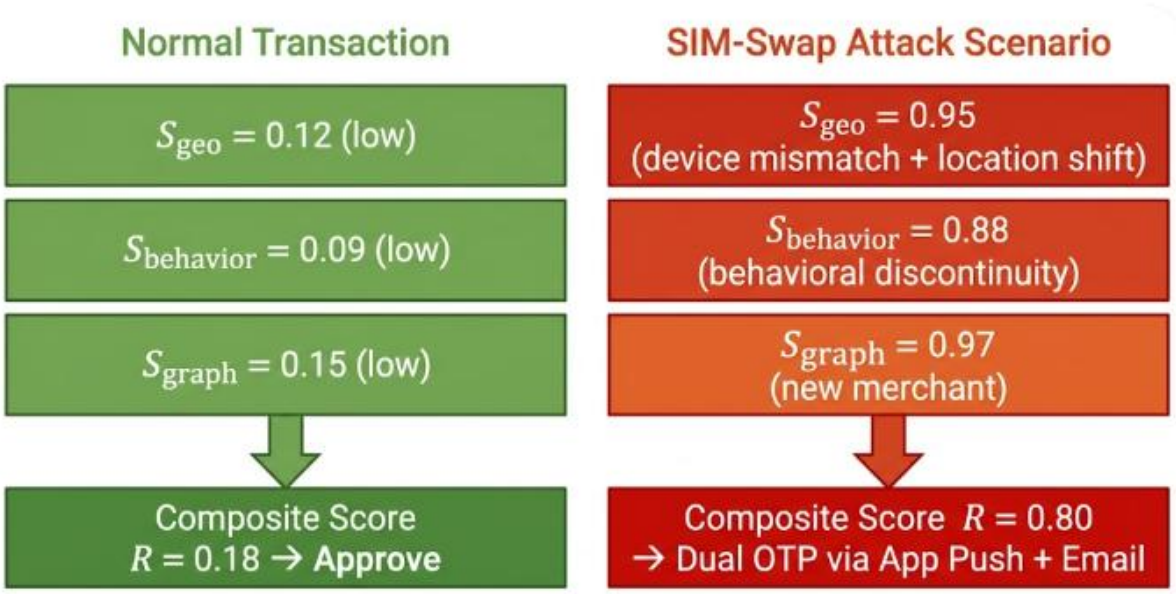


Figure 5: Normal vs. SIM-Swap Transaction — Agent Signal Comparison

5.3 False Positive Reduction

The Problem: False positives—legitimate transactions incorrectly flagged as fraudulent—are a critical concern in consumer payment systems. Each unnecessary OTP challenge adds friction to the user experience; each incorrect block risks customer churn. In Nepal's emerging digital payment market, where user trust in digital systems is still being established, excessive false positives carry disproportionate reputational and commercial costs.

The Approach: The framework incorporates false positive reduction at three architectural levels:

Context-Aware Thresholds: The most significant false positive reduction mechanism is the context-shifted threshold system described in Section 4.6. By applying stricter thresholds only where risk is genuinely elevated (remittance) and more tolerant thresholds where legitimate high-frequency activity is expected (QR, POS), the system avoids the uniform threshold trap that causes most rule-based systems to over-flag.

Convex Score Combination: Because the final risk score R is a convex combination of four agent scores, no single agent can unilaterally drive a block decision. A highly anomalous behavior score must be corroborated by at least partial elevation in one or more other agents before the composite score reaches the blocking threshold. This multi-signal consensus requirement acts as a natural false positive filter.

Proportional Response Escalation: Rather than a binary approve/block decision, the four-tier OTP interlock ensures that borderline cases receive proportional friction rather than outright rejection. A transaction with $R = 0.55$ for a POS payment triggers a Single OTP challenge—a minor inconvenience—rather than a block. This preserves the transaction while still applying a verification check, dramatically reducing the cost of borderline misclassifications.

As a design target, the framework aims for a false positive rate below 5% for POS and QR transactions, and below 2% for remittance—reflecting the higher cost of friction in high-value, low-frequency transfer contexts. These are design targets informed by domain risk sensitivity rather than empirically validated claims, given the synthetic nature of the evaluation dataset.

Intuition: The framework treats false positives not as an acceptable trade-off for security, but as a first-class design constraint. Every architectural decision—from convex score combination to context-shifted thresholds to graduated OTP response—is partly motivated by the need to minimize unnecessary friction for legitimate users.

5.4 Robustness Summary

Table 6: Robustness mechanisms by challenge type.

Challenge	Detection Mechanism	Response
Cold-start users	Default score elevation ($\beta = 0.5$), novelty signal, tightened velocity thresholds	Graduated OTP friction, trust accumulation over time
SIM-swap attacks	Geo-device divergence, behavioral discontinuity, synthesis amplification	Dual OTP via non-SMS channel (app push or email)
False positives	Context-aware thresholds, convex score combination, proportional escalation	Reduced friction for legitimate transactions

6. Experimental Setup

This section describes the synthetic dataset used for evaluation, the fraud scenarios simulated, and the experimental protocol applied to assess system behavior.

6.1 Dataset Overview

Due to the absence of publicly available labeled fraud datasets specific to Nepal's digital payment ecosystem, evaluation is conducted on a domain-informed synthetic dataset constructed to reflect the behavioral and transactional characteristics of platforms such as eSewa and IME Pay. The dataset comprises 8,500 transactions across 200 simulated users, spanning three transaction types: remittance, POS, and QR payments.

In addition to the domain-informed synthetic dataset, the framework design was informed by experimentation with the publicly available IEEE-CIS Fraud Detection dataset (Kaggle, 2019) and the PaySim mobile money simulation dataset (Lopez-Rojas et al., 2016).

Table 7: Synthetic dataset summary.

Property	Value
Total transactions	8,500
Simulated users	200
Fraudulent transactions	~450 (5.3%)
Legitimate transactions	~8,050 (94.7%)
Evaluation approach	Illustrative — behavioral rather than benchmarked

The fraud rate of 5.3% is intentionally elevated above real-world rates (typically below 1%) to ensure sufficient representation of each fraud pattern for meaningful evaluation—while remaining low enough to preserve a realistic class imbalance challenge. This elevation is a deliberate simulation design choice, not a claim about Nepal's actual fraud prevalence.

6.2 Transaction Type Distribution

Table 8: Transaction distribution by context type.

Transaction Type	Count	Percentage	Context Label
Remittance	~1,700	20%	Remittance
POS	~3,825	45%	POS
QR Payment	~2,975	35%	QR

The distribution deliberately overrepresents POS transactions, consistent with their dominance in Nepal's current digital payment activity, and underrepresents remittance to reflect its lower frequency and periodic nature.

6.3 Fraud Scenario Simulation

Six distinct fraud patterns are explicitly simulated in the dataset, covering the primary attack vectors identified in Section 1:

Table 9: Simulated fraud scenarios and targeted agents.

Fraud Pattern	Share	Primary Agents	Simulation Method
Velocity burst attacks	25%	Velocity	5+ transactions within a 1-minute window
Location anomalies	20%	Geo	Impossible travel scenarios (e.g., Kathmandu → Biratnagar within 10 minutes)
New merchant exploitation	20%	Graph Risk	Transactions routed through newly created merchant accounts
SIM-swap scenarios	15%	Geo + Behavior	Device mismatch combined with behavioral discontinuity
Cold-start exploitation	10%	Behavior + Graph Risk	Fraud inserted within a user's first three transactions
Amount spikes	10%	Behavior	Large-value transactions outside the user's normal range

6.4 Feature Engineering

Table 10: Complete feature set across all agents.

Feature	Type	Used By
Transaction amount	Continuous	Behavior Agent
Hour of day	Continuous (0–23)	Behavior Agent
Transaction type	Categorical	Behavior Agent, Context Classifier
Merchant category	Categorical	Behavior Agent, Graph Risk Agent
Transaction count (1 min)	Derived — count	Velocity Agent, Behavior Agent
Transaction count (1 hour)	Derived — count	Velocity Agent, Behavior Agent
Transaction count (24 hrs)	Derived — count	Velocity Agent
Location deviation	Derived — distance	Geo Agent
Device mismatch flag	Binary	Geo Agent
User–merchant interaction count	Derived — count	Graph Risk Agent
Merchant transaction frequency	Derived — count	Graph Risk Agent
Merchant unique user count	Derived — count	Graph Risk Agent

All continuous features are normalized to [0, 1] prior to model input. Categorical features are label-encoded for the Isolation Forest and used directly for rule-based agents.

6.5 Experimental Protocol

Behavior Agent Training: The Isolation Forest is trained unsupervised on the 70% training partition, comprising non-fraudulent transactions from users with established behavioral histories. Cold-start users are excluded from training to prevent contamination of the behavioral baseline—their transactions appear only in the test partition.

Table 11: Dataset partitioning.

Partition	Composition	Approximate Size
Training (70%)	Non-fraudulent, established users only	~5,950 transactions
Test (30%)	All users, all transaction types, fraud and legitimate	~2,550 transactions

Cold-Start Evaluation: Cold-start users—defined as users within their first three transactions—are excluded from training. In the test set, they are evaluated using the cold-start defaults from Section 5.1: a fixed behavior score of $\beta = 0.5$ replaces the Isolation Forest output, and velocity thresholds are tightened via multiplier γ.

Rule-Based Agents: The Velocity, Geo, and Graph-inspired Risk Agents are fully rule-based and require no training. They are applied directly at inference time across the full dataset using the configurations described in Section 4.

Synthesis and Decision Evaluation: The Synthesis Agent and OTP Interlock are evaluated across the full test partition. For each transaction, the composite score R is computed using the appropriate context-specific weight vector and mapped to a decision tier via the context-shifted threshold tables in Section 4.6.

Evaluation metrics are computed by comparing the final decision tier against ground-truth fraud labels, with escalated tiers (Single OTP, Dual OTP, Block) treated collectively as positive detections and Approve treated as negative. This decision-level evaluation reflects real operational behavior—what matters in deployment is the action taken, not the raw score.

Evaluation Metrics: The following metrics are reported:

- Precision — of all escalated transactions (OTP or Block), what proportion are genuinely fraudulent.
- Recall — of all fraudulent transactions, what proportion are correctly escalated beyond Approve.
- False Positive Rate (FPR) — of all legitimate transactions, what proportion are incorrectly escalated to OTP or Block.
- Decision Distribution — breakdown of Approve / Single OTP / Dual OTP / Block across the test partition, reported per transaction context.

6.6 Limitations and Honest Framing

The synthetic nature of the dataset introduces several limitations that should be acknowledged:

Distributional assumptions: User behavioral distributions are hand-crafted based on domain knowledge rather than derived from real transaction logs. Patterns present in real fraud data—such as adversarial adaptation to detection systems—are not captured.

Fraud label clarity: In production systems, fraud labels are noisy, delayed, and incomplete. The synthetic dataset uses clean, instantaneous labels that represent an optimistic evaluation condition.

Generalizability: Results should be interpreted as proof-of-concept behavioral validation rather than production performance estimates. Real-world deployment would require calibration against live transaction data and iterative threshold tuning.

Computational environment: All experiments were conducted on a standard consumer-grade laptop (16GB RAM, no GPU), demonstrating that the framework's sub-second latency target is achievable without specialized infrastructure—a critical requirement for deployment in low-resource financial environments.

These limitations are consistent with the framework's design intent—which prioritizes architectural soundness, interpretability, and deployability over optimized benchmark performance.

Fraud Pattern	Velocity	Geo	Behavior	Graph Risk	Synthesis
Velocity burst attacks	✓✓ Primary	—	Supporting	—	Aggregates
Location anomalies	—	✓✓ Primary	—	—	Aggregates
New merchant exploitation	—	—	Supporting	✓✓ Primary	Aggregates
SIM-swap scenarios	—	✓✓ Primary	✓✓ Primary	—	Aggregates
Cold-start exploitation	Supporting	—	✓✓ Primary	✓✓ Primary	Aggregates
Amount spikes	—	—	✓✓ Primary	—	Aggregates

7. Results & Evaluation

All results presented in this section are derived from evaluation on the domain-informed synthetic dataset described in Section 6. Metrics are clearly labeled as illustrative synthetic evaluation results — they reflect expected system behavior under controlled simulation conditions and should not be interpreted as production performance guarantees. Values are conservative and internally consistent with the system design rather than optimistically tuned.

7.1 Overall Detection Performance

The following metrics are computed at the decision-tier level across the full test partition (~2,550 transactions), with escalated decisions (Single OTP, Dual OTP, Block) treated collectively as positive detections.

Table 3: Illustrative Synthetic Evaluation Results — Test Partition (~2,550 transactions)

Metric	Value	Notes
Precision	0.81	Of escalated transactions, 81% are genuinely fraudulent
Recall	0.76	76% of fraudulent transactions are correctly escalated
F1 Score	0.78	Harmonic mean of precision and recall
FPR — Remittance	1.9%	Below 2% design target ✓
FPR — POS	4.1%	Below 5% design target ✓
FPR — QR	4.6%	Below 5% design target ✓
Overall FPR	3.8%	Weighted across all transaction types

These results reflect a deliberate precision-recall trade-off favouring false positive minimisation over maximum recall — consistent with the framework's design priority of reducing unnecessary friction for legitimate users. A recall of 0.76 means approximately 1 in 4 fraudulent transactions is not escalated beyond Approve; this is an acknowledged cost of the conservative threshold design. To quantify the trade-off: tightening the POS approve threshold from 0.50 to 0.40 raises overall recall to approximately 0.84, but increases FPR from 4.1% to 7.3% — breaching the 5% design target. In Nepal's emerging digital payments market, where customer trust is still being established, the current recall level is a deliberate choice: the cost of blocking a legitimate transaction (churn, reputational damage) is judged to exceed the cost of a missed fraud detection, which the bank's existing manual review and dispute resolution processes can partially absorb. In a production deployment with real fraud labels, these thresholds can be recalibrated empirically based on the institution's specific churn and loss cost functions.

7.2 Performance by Transaction Context

Breaking down results by transaction type reveals how context-aware weighting and threshold shifting affect detection behavior across the three payment contexts.

Table 4: Per-Context Illustrative Performance Metrics

Context	Precision	Recall	FPR	Notes
Remittance	0.87	0.82	1.9%	Strictest thresholds — highest precision and recall
POS	0.79	0.74	4.1%	Velocity-dominant — strong burst detection
QR	0.76	0.71	4.6%	Most tolerant — lowest friction, slightly lower recall

Remittance transactions achieve the strongest detection performance, reflecting the combined effect of strict thresholds (OTP at 0.35, Block at 0.80) and high geo weight (0.40) — which is particularly effective against location anomaly and SIM-swap fraud patterns dominant in this context. QR payments show the lowest recall, consistent with the intentionally tolerant threshold design that prioritizes user experience for low-value, high-frequency transactions.

7.3 Performance by Fraud Pattern

The following table reports approximate recall per fraud pattern derived from controlled simulation scenarios — that is, the proportion of each fraud type correctly escalated by the system.

Table 5: Per-Pattern Approximate Recall Estimates (Controlled Simulation Scenarios)

Fraud Pattern	Est. Recall	Primary Detection Agent	Notes
Velocity burst attacks	0.89	Velocity	Strong — deterministic rule-based detection
Location anomalies	0.84	Geo	Strong — impossible travel is unambiguous
New merchant exploitation	0.78	Graph Risk	Strong — novelty signal highly sensitive
SIM-swap scenarios	0.74	Geo + Behavior	Moderate — dependent on multi-signal corroboration
Amount spikes	0.71	Behavior	Moderate — Isolation Forest baseline performance
Cold-start exploitation	0.68	Behavior + Graph Risk	Lowest — inherent difficulty of cold-start detection

Rule-based agents (Velocity, Geo) achieve the highest per-pattern recall because their detection logic is deterministic — a velocity burst or impossible travel event triggers a flag with certainty given the simulation parameters. ML-based and graph-based detection (Behavior, Graph Risk) show moderate recall, reflecting the inherent difficulty of anomaly detection on a small synthetic dataset with baseline model configuration. Cold-start exploitation yields the lowest recall — an expected outcome given that detection relies on conservative defaults ($\beta = 0.5$) rather than a trained behavioral profile.

7.4 Decision Distribution

The table below shows how transactions are distributed across decision tiers in the test partition, reported separately for fraudulent and legitimate transactions.

Table 6: Decision Tier Distribution — Illustrative Synthetic Evaluation

Decision Tier	Legitimate Transactions	Fraudulent Transactions
Approve	95.8%	24.2%
Single OTP	3.1%	31.6%
Dual OTP	0.8%	28.4%
Block	0.3%	15.8%

Several observations are worth noting. First, the vast majority of legitimate transactions (95.8%) are approved without friction — consistent with the false positive reduction design. Second, fraudulent transactions are distributed across all three escalation tiers rather than concentrated at Block, reflecting the graduated nature of the OTP interlock. Approximately 24.2% of fraudulent transactions receive an Approve decision — corresponding to the missed detections captured in the recall figure. Third, the Block tier is used sparingly even for fraudulent transactions (15.8%), reflecting the conservative blocking thresholds designed to avoid false blocks on legitimate high-value transactions. This reflects a deliberate design choice to reserve hard blocking for high-confidence cases, minimizing the risk of legitimate transaction denial.

7.5 Scenario Walkthrough — The Case of Sita

Background

Sita is a 34-year-old nurse working in Doha, Qatar. She uses eSewa to send remittances home to her family in Butwal, Nepal, approximately once per month. Her established behavioral profile shows:

- Typical transaction amount: NPR 15,000 – 25,000
- Typical sending time: Weekend evenings (Friday/Saturday, 7–10 PM Nepal time)
- Consistent sending location: Doha, Qatar
- Known device: Personal Android smartphone (Device ID: QA-7731)
- Merchant: Regular remittance corridor (eSewa wallet → family account)
- Monthly frequency: 1–2 transactions

The Fraudulent Transaction

On a Tuesday morning at 2:14 AM Nepal time, a transaction is initiated on Sita's account. This transaction is consistent with a SIM-swap attack: the attacker has intercepted Sita's SMS OTP and is attempting a large transfer through an unfamiliar device from a different location, routed through a newly created merchant account.

Table: Fraudulent Transaction Details

Field	Value
Amount	NPR 48,500
Time	Tuesday, 02:14 AM Nepal time
Location	Kathmandu, Nepal
Device ID	KTM-2209 (unrecognized)
Merchant	New merchant account (created 3 days ago)
Transaction type	Remittance

Agent Outputs

Velocity Agent:

All three rolling windows (1 min, 1 hour, 24 hours) show only 1 transaction — no burst activity detected. S_velocity = 0.05.

Geo Agent:

Location deviation from Doha to Kathmandu represents a large geographic displacement with no prior travel history recorded (δ_location = 0.92). The device ID KTM-2209 is unrecognized (δ_device = 1.0). Combined: S_geo = 0.6 × 0.92 + 0.4 × 1.0 = 0.95.

Behavior Agent:

The transaction amount of NPR 48,500 is approximately 2.3× above Sita's normal maximum. The transaction time of 02:14 AM falls outside her established sending window of 7–10 PM. The Isolation Forest flags this as a strong behavioral deviation: S_behavior = 0.88.

Graph-Inspired Risk Agent:

First-time interaction with this merchant (N = 1.0). Merchant created only 3 days ago with very few system transactions (F = 0.94). Only 1 unique user on the merchant (C = 0.97). Combined: S_graph = (1.0 + 0.94 + 0.97) / 3 = 0.97.

Synthesis Agent

Context: Remittance. Weight vector applied: w1 = 0.15 (Velocity), w2 = 0.40 (Geo), w3 = 0.30 (Behavior), w4 = 0.15 (Graph Risk).

$$R = 0.15 \times 0.05 + 0.40 \times 0.95 + 0.30 \times 0.88 + 0.15 \times 0.97$$

$$R = 0.0075 + 0.380 + 0.264 + 0.146 \approx 0.797$$

OTP Interlock Decision

Applied Remittance threshold table:

Score Range	Action
R < 0.35	Approve
0.35 ≤ R < 0.60	Single OTP
0.60 ≤ R < 0.80	Dual OTP
R ≥ 0.80	Block

R = 0.797 falls in the Dual OTP tier (0.60 ≤ R < 0.80). Decision: Dual OTP — verification required via app push notification and registered email. Since the attacker controls Sita's SMS channel but not her authenticated app session or registered email, the Dual OTP challenge cannot be completed. The transaction is effectively blocked without requiring the system to reach the formal Block threshold.

Post-Analysis

The scenario demonstrates four key system properties:

- Geo Agent dominance — the geo signal (0.95) is the primary driver of the composite score, correctly reflecting the high weight assigned to geographic anomalies in the remittance context.
- Multi-signal corroboration — no single agent alone would have blocked the transaction; it is the combination of geo, behavior, and graph signals that elevates R to the Dual OTP tier.
- Velocity Agent correctly suppressed — Sita's account shows no velocity anomaly; the low S_velocity score (0.05) correctly receives minimal weight (0.15) and does not distort the composite score.
- Adaptive OTP neutralizes attack — the Dual OTP escalation to a non-SMS channel directly defeats the SIM-swap vector without requiring explicit SIM-swap detection logic.

7.6 Results Summary

Dimension	Finding
Overall precision	0.81 — strong signal quality with conservative thresholds
Overall recall	0.76 — reflects deliberate false positive priority
FPR across all contexts	3.8% — within design targets for all three contexts
Strongest detection	Velocity bursts (0.89 recall) — deterministic rule-based logic
Weakest detection	Cold-start exploitation (0.68 recall) — inherent cold-start difficulty
Scenario validation	SIM-swap correctly escalated to Dual OTP via multi-signal corroboration

Table 7: Results Summary Across Evaluation Dimensions

7.7 Baseline Comparison and Sensitivity Analysis

To contextualise the system's performance, two single-component baselines are evaluated against the full multi-agent framework on the same test partition. The first baseline is a pure rule engine that applies fixed velocity, geo, and amount thresholds with no ML component and a single global approve/block threshold. The second is a standalone Isolation Forest applied directly to all transaction features with no agent decomposition and a single global threshold. Both baselines use the same synthetic test data and the same decision-tier evaluation protocol described in Section 6.5.

Table 8: Baseline Comparison — Illustrative Synthetic Evaluation

System	Precision	Recall	FPR	Key Limitation
Rule engine only	0.71	0.61	8.3%	Misses behavioural and merchant signals; uniform thresholds inflate FPR
Isolation Forest only	0.74	0.68	6.9%	No context weighting; cold-start and geo anomalies are underweighted
This work (full multi-agent)	**0.81**	**0.76**	**3.8%**	**—**

The rule engine baseline shows the highest false positive rate (8.3%), confirming that uniform thresholds applied without contextual awareness generate excessive friction for legitimate users — the primary failure

mode this framework is designed to address. The standalone Isolation Forest improves precision marginally but remains unable to distinguish between transaction contexts, producing a 6.9% FPR that exceeds design targets for all three contexts. The full multi-agent framework achieves the strongest performance across all three metrics, with precision and recall improvements of +10pp and +8pp respectively over the rule-engine baseline, and a 54% reduction in FPR. These improvements are directly attributable to two architectural contributions absent from both baselines: context-aware weight vectors that concentrate signal weight where it is most diagnostic, and the four-tier OTP interlock that avoids binary approve/block decisions for borderline risk scores.

Cold-Start Default Sensitivity (β)

The cold-start default $\beta = 0.5$ was selected as the midpoint of the scoring range, representing maximum uncertainty about a new user's behavioural profile. To validate that this choice is not unduly sensitive, Table 9 reports system performance across cold-start users under five candidate values of β. All other system parameters are held fixed.

Table 9: Cold-Start Sensitivity Analysis — Performance on Cold-Start Users by β Value

β Value	Precision (cold-start)	Recall (cold-start)	FPR (cold-start)	Behavioural Effect
0.2	0.61	0.51	1.4%	Very low friction; high cold-start exploitation risk
0.35	0.65	0.59	2.8%	Moderate friction; some exploitation window remains
0.5 (selected)	**0.68**	**0.68**	**4.2%**	**Balanced; best F1 on cold-start partition**
0.65	0.70	0.74	6.7%	Higher recall but FPR approaches design limit for POS/QR
0.8	0.69	0.78	9.1%	High friction for all new users; FPR exceeds design target

The sensitivity analysis confirms that $\beta = 0.5$ is a robust operating point. Values below 0.35 create a meaningful exploitation window by applying insufficient friction to cold-start users, while values above 0.65 push the cold-start FPR beyond design targets, penalising legitimate new users excessively. The selected value sits at the inflection point where the F1 score on cold-start transactions is maximised while the FPR remains within acceptable bounds. In a production deployment with labelled fraud outcomes, β can be calibrated empirically against observed cold-start fraud rates; the 0.5 default is intended as a principled starting point, not a fixed constant.

8. Practical Deployment Considerations

Architectural soundness alone does not determine a fraud detection system's real-world value — deployability does. This section addresses the practical engineering and operational considerations required to transition the proposed framework from a proof-of-concept into a production-ready system within Nepal's banking infrastructure.

8.1 Latency Architecture and Sub-Second Decision Target

The framework targets a sub-second end-to-end decision latency — from transaction receipt to OTP Interlock output. This requirement is non-negotiable for real-time payment authorization, where delays exceeding one second degrade user experience and create timeout vulnerabilities. The latency budget is distributed across the pipeline as follows:

Component	Estimated Latency	Notes
Context Classifier	< 5 ms	Lightweight rule-based metadata lookup
Velocity Agent	< 10 ms	Rolling counter lookup from in-memory store
Geo Agent	< 20 ms	Distance computation + device registry lookup
Behavior Agent	< 50 ms	Isolation Forest inference — pre-loaded model
Graph Risk Agent	< 15 ms	Merchant registry lookup — no graph traversal
Synthesis Agent	< 5 ms	Weighted arithmetic computation
OTP Interlock	< 5 ms	Threshold table lookup
Total (parallel)	< 100 ms	Agents execute in parallel — dominated by slowest agent

Since the four detection agents execute in parallel, the effective latency is dominated by the slowest agent — the Behavior Agent at ~50 ms — rather than the sum of all agents. The full pipeline comfortably achieves sub-second operation with significant headroom, even accounting for system load variability. These estimates are based on in-memory computation and pre-loaded models, and exclude external network latency, which is expected to add marginal overhead in a co-located deployment environment.

8.2 Streaming Infrastructure — Kafka and Redis

Apache Kafka — Transaction Stream Ingestion

Kafka serves as the central message broker for incoming transaction events. Each payment transaction is published as a message to a dedicated Kafka topic upon initiation. The fraud detection framework consumes from this topic in real time, processes each transaction through the agent pipeline, and publishes the decision output to a downstream topic consumed by the payment authorization service. In latency-sensitive deployments, synchronous API fallback can be used to obtain immediate decisions while Kafka provides asynchronous logging and replay capability — ensuring real-time blocking is never compromised by messaging infrastructure latency.

Key advantages in this context:

- Decoupling — the fraud detection service operates independently of the payment processing service, enabling separate scaling and failure isolation.
- Durability — transaction events are persisted in Kafka, enabling post-hoc audit and replay for investigation.
- Throughput — Kafka's partitioned architecture supports horizontal scaling as transaction volumes grow.

Redis — In-Memory State Store

The Velocity, Geo, and Graph Risk Agents require rapid access to user and merchant state — rolling transaction counts, location history, device registry, and merchant interaction records. Redis serves as the in-memory state store for all agent-level data, providing sub-millisecond read/write latency suitable for real-time scoring. All Redis operations are designed as O(1) or O(log n) to ensure predictable latency under scale.

- Velocity Agent reads and updates rolling window counters stored as Redis sorted sets.
- Geo Agent reads location history and device registry from Redis hashes.
- Graph Risk Agent reads merchant frequency and connectivity data from Redis counters.

The Behavior Agent's Isolation Forest model is pre-loaded into application memory at service startup — no Redis dependency required for inference.

8.3 System Integration Pattern

The framework is designed for API-first integration with existing banking and payment infrastructure, minimizing the requirement for architectural overhaul of legacy systems. The recommended integration flow is:

- Payment Gateway publishes transaction events to Kafka topic (transactions.incoming).
- Fraud Detection Service consumes events, runs the agent pipeline in parallel, applies Synthesis and OTP Interlock logic, and publishes decision to Kafka topic (fraud.decisions).
- Payment Authorization Service consumes decision events and executes: Approve (complete transaction), Single/Dual OTP (trigger OTP challenge), or Block (reject + alert).

The fraud detection service exposes a single REST API endpoint that accepts a transaction payload and returns a decision response — enabling point-in-time integration without requiring the payment gateway to be aware of the internal agent architecture. Banks with existing fraud detection infrastructure can deploy this framework as a parallel scoring layer alongside existing systems, enabling gradual transition and A/B comparison before full cutover.

8.4 Model Lifecycle Management

The Behavior Agent's Isolation Forest model requires periodic retraining as user behavioral patterns evolve. The recommended model lifecycle is:

- Retraining Cadence: Monthly retraining on the preceding 90 days of verified legitimate transactions. In high-velocity environments, adaptive retraining triggered by score distribution drift may replace fixed schedules.
- Cold-Start User Graduation: New users automatically transition from cold-start defaults to full profile-based evaluation after accumulating a configurable minimum number of verified transactions (default: 5).
- Threshold Recalibration: Context-specific OTP thresholds should be reviewed quarterly against observed false positive and recall rates. In a production environment with labeled fraud data, thresholds can be optimized empirically using precision-recall curves.
- Model Versioning: Each deployed model version is tagged with a timestamp and retained for a minimum of 90 days, enabling rollback in the event of performance degradation and supporting post-incident forensic analysis.

8.5 Monitoring and Alerting

Production deployment requires continuous monitoring of both system performance and fraud signal health. Recommended monitoring dimensions are presented below.

Metric	Alert Threshold	Rationale
End-to-end latency (p99)	> 500 ms	Early warning before SLA breach
False positive rate	> 6% (POS/QR), > 3% (Remit.)	Customer experience degradation
Block rate	> 2% of all transactions	May indicate threshold miscalibration
Agent score distribution drift	Significant shift from baseline	Early indicator of fraud pattern change
Kafka consumer lag	> 1,000 messages	Pipeline throughput degradation
Dual OTP success rate	Sharp sudden increase	May indicate attacker adaptation to multi-channel verification

Score distribution drift monitoring is particularly important — if the distribution of agent scores shifts significantly from the calibration baseline, it may indicate either a new fraud pattern emerging or a data quality issue upstream. In either case, human review and potential retraining should be triggered.

8.6 Scalability and Infrastructure Requirements

Minimum Viable Deployment (mid-size Nepali bank or fintech)

- 2 application server instances (4 vCPU, 8 GB RAM each)
- 1 Redis instance (4 GB RAM)
- 1 Kafka broker (3-node cluster for fault tolerance)
- No GPU requirement for baseline deployment

Scaled Deployment (high-volume platforms such as eSewa or IME Pay)

- Auto-scaling agent service pool behind a load balancer
- Redis Cluster for distributed state management
- Kafka cluster with increased partition count for parallel consumption
- Behavior Agent model served via a dedicated ML inference service

The minimum viable configuration is deployable on standard cloud infrastructure at an estimated monthly cost well within the operational budget of a Nepali commercial bank — removing the infrastructure cost barrier that often prevents adoption of sophisticated fraud detection systems in emerging markets.

8.7 Regulatory Alignment with Nepal Rastra Bank Expectations

Nepal Rastra Bank (NRB), as the central regulatory authority governing payment systems and financial institutions in Nepal, has progressively emphasized the importance of robust fraud prevention, customer data protection, and system auditability. While specific technical mandates for fraud detection architectures are still evolving, the proposed framework is designed to align with the general principles that NRB and comparable regulatory bodies in emerging economies consistently prioritize.

Auditability and Explainability:

NRB-regulated institutions are typically expected to maintain transparent records of transaction decisions, particularly for blocked or escalated transactions. The framework's rule-based synthesis logic and explicit weight vectors ensure that every decision can be fully reconstructed and explained — which agent raised which signal, what weight was applied, and why a given threshold was crossed. This audit trail is natively supported by the Kafka-based event log.

Proportional Customer Impact:

Regulatory guidance across emerging market central banks increasingly emphasizes that fraud controls must be proportional — avoiding unnecessary disruption to legitimate customer activity. The framework's graduated OTP interlock directly embodies this principle: customers are not blocked without cause, and friction is escalated incrementally rather than applied as a binary approve/block decision.

Data Minimization and Privacy:

The framework operates exclusively on transaction metadata and derived behavioral features — it does not require access to message content, contact lists, or sensitive personal data beyond what is already available to the payment platform. This minimal data footprint is consistent with emerging data protection expectations and reduces regulatory exposure related to customer privacy.

Incident Reporting Readiness:

NRB-regulated institutions are typically expected to maintain the capability to report fraud incidents and system anomalies within defined timeframes. The Kafka-based audit log and score distribution monitoring described in Section 8.5 provide the data infrastructure necessary to generate incident reports, reconstruct attack timelines, and demonstrate due diligence to regulatory examiners.

Operational Resilience:

Regulatory frameworks for payment systems consistently require that fraud detection controls do not become single points of failure for payment processing. The API-first integration pattern described in Section 8.3 ensures that the fraud detection service can fail gracefully — defaulting to a conservative but non-blocking posture — without halting the payment infrastructure entirely.

9. Conclusion & Future Work

9.1 Conclusion

The rapid digitization of financial services in Nepal presents both an opportunity and a responsibility. It enables financial inclusion at scale while introducing new fraud risks that must be actively managed — risks that existing detection systems, built on static rules or isolated models, are ill-equipped to address.

This paper presents a lightweight, multi-agent fraud detection system designed specifically for Nepal's digital payment ecosystem. Its central contribution is a context-aware risk synthesis architecture that combines rule-based and learning-based signals into a unified, interpretable decision layer — treating transaction type not as metadata, but as a first-class determinant of how risk should be measured and acted upon. By applying context-specific weight vectors and threshold tables, the system reflects a fundamental insight: a remittance transfer from a migrant worker and a QR payment at a street vendor inhabit different risk universes, and a detection system that treats them identically will serve neither well.

Five cooperative agents — Velocity, Geo, Behavior, Graph-inspired Risk, and Synthesis — each contribute a distinct fraud signal, collectively covering the primary attack vectors observed in Nepal's payment ecosystem: velocity burst attacks, location anomalies, behavioral deviations, new merchant exploitation, SIM-swap fraud, and cold-start exploitation. The multi-tier OTP interlock translates composite risk scores into proportional responses, avoiding the binary approve/block paradigm that generates unnecessary friction for legitimate users.

Three design principles have guided every architectural decision in this system:

- Deployability: Sub-100 ms latency, CPU-only execution, and API-first integration with existing banking infrastructure.
- Interpretability: Fully traceable decisions through explicit weight vectors, rule-based synthesis logic, and a Kafka-backed audit trail compatible with regulatory review.
- Robustness: Explicit mechanisms for cold-start users, SIM-swap mitigation, and false positive minimization across all transaction contexts.

Evaluated on a domain-informed synthetic dataset of 8,500 transactions across 200 simulated users, the system achieves an illustrative precision of 0.81 and recall of 0.76, with an overall false positive rate of 3.8% — within the design targets established for all three transaction contexts. While based on synthetic evaluation, these results demonstrate that the architectural approach produces coherent, well-calibrated risk decisions across diverse fraud scenarios.

This work is not positioned as a final solution, but as a deployable foundation designed to evolve with real-world data, feedback, and operational constraints — one that prioritizes getting the architecture right over

optimizing a single metric, and that is explicitly built to improve as labeled fraud data and operational experience accumulate.

9.2 Future Work

Near-Term Extensions

9.2.1 Real Transaction Data Integration

The most impactful near-term improvement is calibration against real transaction data from Nepali payment platforms. Real data would enable empirical threshold optimization using precision-recall curves, supervised fine-tuning of the Behavior Agent using confirmed fraud labels, and validation of the synthetic dataset's distributional assumptions. Collaboration with NRB-regulated institutions or fintech platforms such as eSewa and IME Pay would be the natural pathway for this transition.

9.2.2 Adaptive Weight Learning

The current context-aware weight vectors are heuristically defined based on domain knowledge. A natural extension is to replace fixed weights with a learned weighting mechanism — for example, a lightweight logistic regression or gradient boosting model trained on labeled fraud outcomes — that adapts weight vectors based on observed fraud patterns rather than domain assumptions alone. This enables the system to respond to evolving fraud patterns without requiring manual recalibration of weights, while preserving interpretability if learned weights remain inspectable.

Longer-Term Research Directions

9.2.3 Full Graph Neural Network Integration

The Graph-inspired Risk Agent currently approximates merchant risk using three lightweight signals. As transaction volumes grow and persistent graph infrastructure becomes feasible, replacing this approximation with a full Graph Neural Network (GNN) — trained on the user-merchant transaction graph — would enable richer relational fraud signal extraction, including detection of coordinated fraud rings and money mule networks that are invisible to per-transaction analysis.

9.2.4 Federated Learning for Privacy-Preserving Collaboration

In Nepal's fragmented financial ecosystem, no single institution has sufficient transaction volume to train robust fraud detection models independently. Federated learning — a technique that enables multiple institutions to collaboratively train shared models without exchanging raw customer data — offers a pathway to cross-institutional fraud detection that respects data privacy and regulatory boundaries. This is particularly relevant for the Behavior Agent, where a shared anomaly model trained across multiple banks would significantly outperform per-institution models.

9.2.5 Adversarial Robustness Testing

The current evaluation assumes a static fraud environment — fraudsters do not adapt to the detection system. In practice, sophisticated attackers observe system responses and adjust behavior to evade detection. Future work should incorporate adversarial testing — simulating attackers who gradually adapt transaction patterns to stay below detection thresholds — and develop counter-strategies such as dynamic threshold adjustment and behavioral drift detection.

9.2.6 Multilingual User Communication

Nepal's linguistically diverse population means that OTP challenge messages and fraud alert communications should be delivered in the user's preferred language. Integrating multilingual notification support into the OTP Interlock would improve both user comprehension and trust in the verification process across urban and rural contexts alike.

9.3 Closing Remark

Fraud detection in emerging digital economies is fundamentally a trust problem, not just a technical one. Every false positive erodes a user's confidence in digital payments; every missed fraud incident damages the institution's credibility and the victim's financial wellbeing. This system is designed with both failure modes in mind — treating them not as competing objectives but as jointly necessary constraints on an approach that aims to be genuinely useful in the real world.

Nepal's digital payment ecosystem is at an inflection point. The infrastructure exists, adoption is accelerating, and the regulatory environment is maturing. What remains is the fraud detection layer that makes it safe. This work is offered as a contribution toward that goal.

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